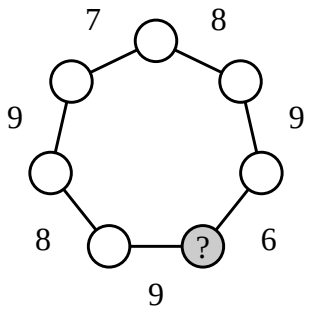
Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q23 (v1)

Sums between neighbours in a ring: guess the number in one circle, go around the ring and check that it closes

How to think about it: sums in a ring

Each number between two circles is the sum of its two neighbours. If you know **one** circle, you know the next one: next = sum – this one. So:

- **1. Guess** a number for the question mark.
- **2. Go around** the ring, circle by circle: next = sum – this one.
- **3. Check** when you come back: do you get your guess again? If not, try another guess.



The contest question

Start at the question mark and go clockwise. The sums on the way are 9, 8, 9, 7, 8, 9, and the last sum, 6, brings you back to the question mark:

guess for ?	going around	back at ?	works?
3	$3 \to 6 \to 2 \to 7 \to 0 \to 8 \to 1$	5	no
4	$4 \to 5 \to 3 \to 6 \to 1 \to 7 \to 2$	4	**yes**
5	$5 \to 4 \to 4 \to 5 \to 2 \to 6 \to 3$	3	no

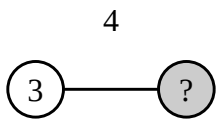
Only the guess **4** comes back to itself. And the numbers are 1 to 7, each once.

Watch out

- Do not split one sum in half (6 = 3 + 3): the two neighbours are different numbers.
- Trick for later: add all the sums. Every circle is counted twice, so the sum of all the circles is half of it.

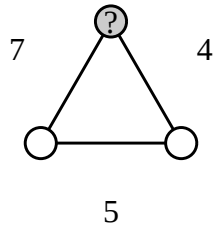
Warm-up: small steps first (no points)

W1. The number between the circles is their sum. Which number is in the circle with the question mark?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 7

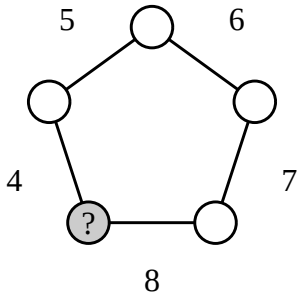
W2. The number between two circles is the sum of these two circles. Which number is in the circle with the question mark? Guess, go around, and check.



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

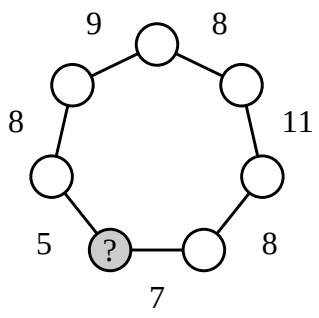
Part A: Each question is worth 3 points

1. Lena writes the numbers from 1 to 5 into the circles, each number exactly once. When she adds the numbers of two circles next to each other, she gets the number written between them. Which number is in the circle with the question mark?



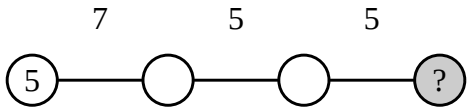
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

2. Lena writes the numbers from 1 to 7 into the circles, each number exactly once. When she adds the numbers of two circles next to each other, she gets the number written between them. Which number is in the circle with the question mark?



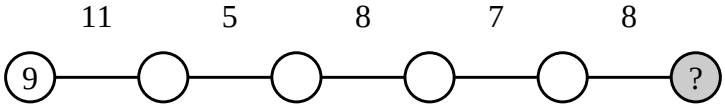
A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

3. The number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 4 ***E.*** 5

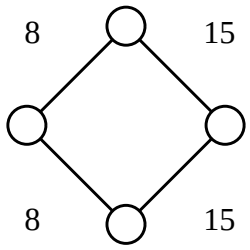
4. The number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

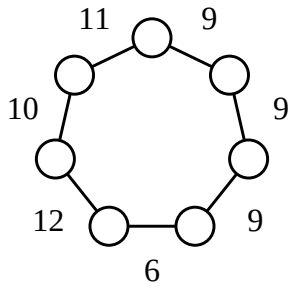
Part B: Each question is worth 4 points

5. Each number between two circles is the sum of the numbers in these two circles. What do the numbers in all the circles add up to?



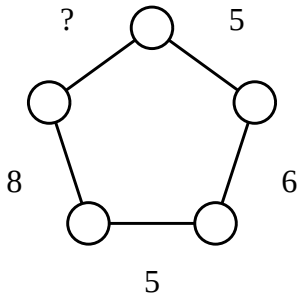
A. 18 ***B.*** 23 ***C.*** 30 ***D.*** 35 ***E.*** 46

6. Each number between two circles is the sum of the numbers in these two circles. What do the numbers in all the circles add up to?



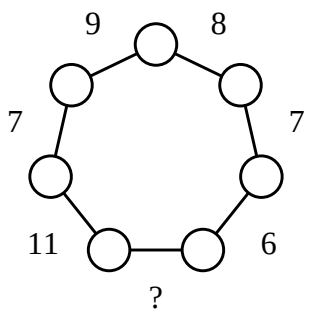
A. 33 ***B.*** 37 ***C.*** 40 ***D.*** 50 ***E.*** 66

7. The numbers from 1 to 5 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number belongs where the question mark is?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

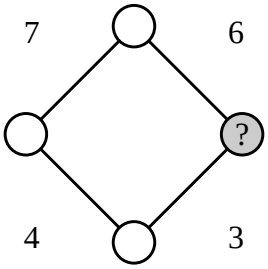
8. The numbers from 1 to 7 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number belongs where the question mark is?



A. 3 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

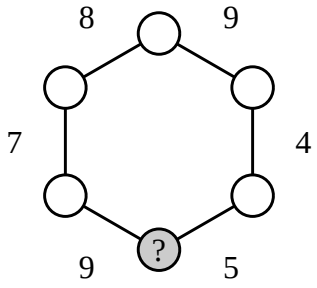
Part C: Each question is worth 5 points

9. The numbers from 1 to 4 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

10. The numbers from 1 to 6 are written into the circles, each exactly once. Each number between two circles is the sum of the numbers in these two circles. Which number is in the circle with the question mark?



A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

Answer Key

W1.	**B**	**W2.**	**C**	**1.**	**C**	**2.**	**A**	**3.**	**C**	**4.**	**E**
5.	**B**	**6.**	**A**	**7.**	**E**	**8.**	**E**	**9.**	**A**	**10.**	**A**

Solutions

W1. B (1) — 4 – 3 = 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · w_next]

W2. C (3) — Only 3 works: 3 → 1 → 4 → 3. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · w_ring_closes]

1. C (3) — Guess a number for the question mark and go around the ring, using the sums: only 3 works all the way round (3, 1, 4, 2, 5). [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · odd_ring]

2. A (4) — Guess a number for the question mark and go around the ring, using the sums: only 4 works all the way round (4, 1, 7, 2, 6, 5, 3). [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · odd_ring]

3. C (2) — Go step by step from the left: 7 – 5 = 2, 5 – 2 = 3, 5 – 3 = 2. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · in_a_row]

4. E (6) — Go step by step from the left: 11 – 9 = 2, 5 – 2 = 3, 8 – 3 = 5, 7 – 5 = 2, 8 – 2 = 6. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · in_a_row]

5. B (23) — Add all the numbers between the circles: 15 + 15 + 8 + 8 = 46. Every circle is in two of these sums, so all the circles together make 46 ÷ 2 = 23. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · all_together]

6. A (33) — Add all the numbers between the circles: 9 + 9 + 9 + 6 + 12 + 10 + 11 = 66. Every circle is in two of these sums, so all the circles together make 66 ÷ 2 = 33. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · all_together]

7. E (6) — 1 + 2 + ... + 5 = 15. Every circle is in two sums, so all the sums together make 2 × 15 = 30. The shown sums make 24, so the missing one is 30 – 24 = 6. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · hidden_sum]

8. E (8) — 1 + 2 + ... + 7 = 28. Every circle is in two sums, so all the sums together make 2 × 28 = 56. The shown sums make 48, so the missing one is 56 – 48 = 8. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · hidden_sum]

9. A (2) — With 4 circles every guess closes the ring, so check something else: the numbers must be 1 to 4, each once. Only 2 gives 2, 1, 3, 4. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · even_ring]

10. A (4) — With 6 circles every guess closes the ring, so check something else: the numbers must be 1 to 6, each once. Only 4 gives 4, 5, 2, 6, 3, 1. [set-01/by_level/Grade_03-04_Ecolier/2023 Q23 · even_ring]